

Technical Description

XBandDigitalModule_20x20 - Technical Description

Document Information

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1. Executive Summary

The XBandDigitalModule_20x20 is a 20cm x 20cm space-grade electronic module for direct RF sampling of X-band signals (8-12 GHz) with 500 MHz instantaneous bandwidth. The module implements a complete s

2. System Architecture

2.1 Signal Chain

RF Input (8-12 GHz, SMA) -> BPF -> LNA -> Balun (TRF0208-SP) -> ADC (ADC12DJ5200-SP) -> JESD204C (16 lanes @ 16 Gbps) -> FPGA (XQRCV1902) -> DDC (8 channels) -> FIFO Buffer -> SpaceWire (2 links @ 100 Mbps)
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2.2 Key Features

Feature	Specification
Input Frequency	8-12 GHz (X-ban
Signal Bandwidt	500 MHz
ADC Resolution	12-bit
Sample Rate	10.4 GSPS
Clock Jitter	<100 fs RMS
DDC Channels	8 independent
Data Output	SpaceWire 2x 10
Power	<100W
Dimensions	200mm x 200mm x
Radiation	100+ krad(Si) T

3. Component Selection

3.1 Critical Components

Component	Part Number	Function	Rationale
ADC	ADC12DJ5200-SP	10.4 GSPS sampl	Only space-grad
FPGA	XQRCV1902	Digital process	44 GTY @ 26.5 G
Clock Synth	LMX2615-SP	10.4 GHz clock	45 fs jitter, Q
Clock Dist	LMK04832-SP	Jitter cleaning	JESD204B/C SYSR
Balun	TRF0208-SP	SE to Diff	DC-8 GHz, match
Buck	TPS7H5002-SP	FPGA core power	Rad-hard, GaN-o
LD0	TPS7H1111-SP	ADC analog powe	1.71 μVRMS ultr
Sequencer	TPS7H3014-SP	Power sequencin	4-channel, prog

7. VHDL Architecture

7.1 Module Hierarchy

```
top_module
+-- jesd204c_receiver (16 lanes, 64B/66B)
+-- ddc_chain (8 channels, NCO + CIC + FIR)
+-- fifo_buffer (4K x 32-bit, async)
+-- error_handler (EDAC, fault logging)
+-- clock_manager (SPI config for LMK04832-SP)
+-- spacewire_interface (2 links, 100 Mbps)
+-- axi4_lite_slave (register interface)
```

7.2 Register Map Summary

Address	Name	Description
0x0000	SYS_STATUS	System status a
0x0004	SYS_CTRL	System control
0x0040	ADC_CTRL	ADC configurati
0x0080	FPGA_TEMP	FPGA temperatur
0x00C0	POWER_STATUS	Power rail stat
0x0100	CLOCK_STATUS	Clock lock stat
0x0200-0x02FF	DDC_CTRL	DDC configurati
0x0300-0x03FF	SPW_CTRL	SpaceWire confi

8. Risk Assessment

Risk	Probability	Impact	Mitigation
Clock jitter >	Low	High	LMX2615-SP (45
JESD204C lane m	Low	Medium	GTY @ 26.5 Gbps
Power sequencin	Medium	High	TPS7H3014-SP pr
TID accumulatio	Low	Medium	100+ krad compo
Thermal hotspot	Medium	Medium	Thermal vias, c

9. Test Approach

9.1 Test Points

TP	Signal	Location
TP1	VIN (4.5V)	Power input
TP2	VCCINT (1.0V)	FPGA core
TP3	VCCAUX (1.8V)	FPGA aux
TP4	VCC0 (3.3V)	I/O
TP5	AVDD (1.0V)	ADC analog
TP6	DVDD (1.8V)	ADC digital
TP7	CLK (10.4 GHz)	Clock output
TP8	SYSREF	JESD sync
TP9	JESD Lane 0	Data monitor
TP10	ERROR	System error

9.2 Test Procedures

1. Power-on test: Verify all rails within spec
2. 3.

10. Conclusion

The XBandDigitalModule_20x20 design is complete and ready for detailed implementation. All critical components have been

selected with adequate margins for space operation. The direct RF sampling arch
